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Original Article

A preliminary quantitative analysis of REM sleep chin EMG in Parkinson's disease with or without REM sleep behavior disorder

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ABSTRACT

Objectives: To compare REM sleep chin EMG quantitative features between Parkinson's disease (PD) patients with or without REM sleep behavior disorder (RBD).
Subjects and methods: Twenty-seven consecutive PD patients (mean age 67.9 years) and 19 normal controls (mean age 67.5 years) were enrolled. Detailed clinical, laboratory, and polysomnographic studies were obtained in all participants and characteristics of chin electromyographic amplitude during rapid eye movements sleep were analyzed by means of an automatic quantitative approach (Atonia Index).
Results: Sixteen of the 27 patients were affected by RBD. An Atonia Index below 0.90 showed high sensitivity (0.938) and specificity (0.909) for the diagnosis of RBD within the group of PD patients.
Conclusion: This study recommends the Atonia Index as an objective measure to support and aid the diagnosis of RBD in PD.

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1. Introduction

REM sleep behavior disorder (RBD) is a REM sleep parasomnia characterized by dream-enacting behavior associated with altered dream content (mainly violent, fearful dreams) that can herald by many years other neurodegenerative diseases, mainly synucleinopathies [1,2]. The current definition of the main polysomnographic feature of RBD REM sleep without atonia (RSWA) has no quantitative characterization [3], thus the diagnosis is essentially based on clinical data and corroborated by the visual inspection of nocturnal polysomnographic (PSG) recordings.

We have recently described an automatic quantitative method able to characterize EMG patterns during REM sleep [4,5] which has been validated with respect to the visual scoring of loss of atonia according to Lapierre and Montplaisir [4,6]. The application of this new method disclosed differences across the life span [7], between normal controls and patients with RBD [4,5], and also between nosologically different groups of patients such as multiple system atrophy [4] or narcolepsy [8], who seem to present with different patterns of REM sleep-related motor disturbances. RBD is also frequently associated with clinically diagnosed Parkinson's disease (PD) [9,10]. So far only one study has applied automated analysis of REM EMG in PD patients [11]; however, results were only reported for the complete and mixed sample of patients with different neurodegenerative diseases.

For this reason, the aim of this study was to characterize RSWA in patients with PD by means of our quantitative approach. We also evaluated the sensitivity and specificity of the REM sleep Atonia Index with respect to clinical RBD diagnosis, which eventually may provide an objective parameter to be used as a supportive criterion for the clinical diagnosis of RBD in patients with PD.

2. Subjects and methods

2.1. Subjects

Twenty-seven consecutive PD patients attending the Department of Neurology I.C. of the Oasi Institute (Troina, Italy) or the Sleep Disorder Center of the Department of Neurological Sciences of the University of Bologna (Italy) were enrolled in the study (17 men and 10 women, mean age 67.9 years, 7.45 S.D.). All patients fulfilled the United Kingdom Parkinson's Disease Society Brain Bank clinical diagnostic criteria for PD [12]. Exclusion criteria were: presence of other neurological diseases, Mini-Mental State Examination score below 20 [13], presence of a psychiatric disease according to the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) [14], and a PSG study showing apnea/hypopnea index above 10 per hour. The presence of RLS was assessed by means

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of the four criteria of the International RLS Study Group [15] but did not cause the exclusion of the patients if they were under dopaminergic treatment.

Eight patients had unilateral involvement only (Hoehn & Yahr [H&Y] stage 1) [16], six subjects had unilateral and axial involvement (H&Y stage 1.5), six had bilateral symptoms without impairment of balance (H&Y stage 2), three patients had mild bilateral disease with recovery on pull test (H&Y stage 2.5), and four patients had balance impairment and moderate disease but were physically independent (H&Y stage 3). Mean disease duration was 4.3 years (3.89 S.D., range 0.5–16 years). Two patients had not yet started any drug therapy for PD; the remaining patients were taking *L*-Dopa monotherapy, dopamine-agonist alone, a combination of the two, or other combinations of *L*-Dopa, dopamine agonists, and other anti-PD drugs (amantadine, selegiline, etc.). In order to carry out comparisons among patients on different dopaminergic treatments, drug dosages were converted to *L*-Dopa Dosage Equivalents (LED), according to Tomlinson et al. [17].

A careful diagnosis of RBD was carried out based on the International Classification of Sleep Disorders – 2nd ed. [3] criteria for RBD and including the presence of REM sleep without atonia, sleep related injurious-disruptive behaviors by history or abnormal sleep behaviors documented during polysomnographic monitoring, and absence of EEG epileptiform activity during REM sleep.

The control group was comprised of 19 age-matched normal controls (seven men and 12 women, mean age 67.5 years, 7.28 S.D.). The exclusion criteria for the control group were the same as described for PD patients; additionally, the presence of subjective sleep complaints (insomnia, daytime sleepiness, restless legs syndrome, REM sleep behavior disorder symptoms, snoring, or witnessed apnea) was also ruled out. None of the controls were taking hypnotics or benzodiazepines.

This study was approved by the local ethics committee and all subjects provided informed consent according to the Declaration of Helsinki before entering the study.

2.2. Polysomnographic sleep recordings

Each subject underwent a full polysomnographic night recording, after an adaptation night, carried out in a standard sound-attenuated (noise level to a maximum of 30 dB nHL) sleep laboratory. Subjects were not allowed to have beverages containing caffeine during the afternoon preceding the recording and were allowed to sleep until their spontaneous awakening in the morning.

The following parameters were included in the polysomnographic study: EEG (at least three channels, one frontal, one central and one occipital, referred to the contralateral earlobe); electrooculogram (electrodes placed 1 cm above the right outer cantus and 1 cm below the left outer cantus and referred to A1), electromyogram (EMG) of the submentalis muscle, EMG of the right and left tibialis anterior muscles (bipolar derivations with two electrodes placed 3 cm apart on the belly of the anterior tibialis muscle of each leg), and ECG (one derivation). The sleep respiratory pattern of each patient was assessed by means of oral and nasal airflow (thermistor or nasal pressure cannula), thoracic and abdominal respiratory effort (strain gauge), and oxygen saturation (pulse-oxymetry) in a previous recording (within one week) or during the study night. Sleep signals were sampled at 200 or 256 Hz and stored on hard disk in European data format for further analysis. Periodic leg movements during sleep (PLMS) were analyzed according to the World Association of Sleep Medicine-International Restless Legs Syndrome Study Group and the American Academy of Sleep Medicine criteria [18,19].

2.3. Quantification of the submentalis muscle EMG amplitude

For the computer quantitative analysis we used an established automatic scoring algorithm [4,5]. The submentalis muscle EMG signal was digitally band-pass filtered at 10–100 Hz, with a notch filter at 50 Hz, and rectified. Subsequently, each REM sleep epoch included in the analysis was divided into 30 1-s mini-epochs. The average amplitude of the rectified submentalis muscle EMG signal was then obtained for each mini-epoch. After a noise reduction procedure [5], the values of the submentalis muscle EMG signal amplitude in each 1-s mini-epoch were used to compute the percentage of values in the following 20 amplitude (amp) classes (expressed in μV): $\text{amp} \leqslant 1$, $1 < \text{amp} \leqslant 2, \ldots,$ $18 < \text{amp} \leqslant 19$, $\text{amp} > 19$. Muscle atonia is expected to be reflected by high values of the first class ($\text{amp} \leqslant 1$), while phasic and tonic activations are expected to increase the value of the other classes [4,5,8]. As proposed in previous studies, an index summarizing in a single value the degree of preponderance of the first class was used in REM sleep:

$$\text{Atonia Index} = \text{amp} \leq 1/(100 - 1 < \text{amp} \leq 2).$$

Mathematically, this index can vary from zero (absence of mini-epochs with $\text{amp} \leqslant 1$), i.e., complete absence of EMG atonia, to one (all mini-epochs with $\text{amp} \leqslant 1$) or stable EMG atonia in the epoch.

In addition, sequences of consecutive mini-epochs with values >2 μV were counted as movements. The algorithm was run blind to the condition of the subject, even though no manual modifications of the parameters was possible.

2.4. Statistical analysis

For the statistical analysis all comparisons were performed by means of the nonparametric Mann–Whitney test for unpaired datasets or the Kruskal–Wallis ANOVA, as appropriate. However, because of the relatively limited number of subjects available, and to rule out possible type II errors, we also calculated effect sizes using Cohen's *d* value [20]. Cohen's *d* is defined as the difference between two means divided by their pooled standard deviation. According to Cohen, 0.2 is indicative of a small effect, 0.5 a medium, and 0.8 of a large effect size. The chi-square test was used for the comparison of frequencies. The commercially available Statistica software package (StatSoft, Inc., 2001. STATISTICA data analysis software system, version 6. www.statsoft.com) was used. Differences were considered significant when they were below the $p < 0.05$ level.

3. Results

3.1. Comparison between normal controls and PD patients

The comparison between the sleep architecture parameters obtained in normal controls and PD patients is reported in Table 1. As expected, in PD patients several features of disrupted sleep were observed with significantly shorter time in bed, sleep period time, and total sleep time; sleep efficiency was also significantly reduced in patients as a consequence of increased wake after sleep onset and a significant reduction of sleep stage 2. Effect sizes were large for these parameters and confirmed results of the statistical comparisons. In addition, effect sizes also suggested that the decrease in the percentage of REM sleep might be significant with a larger sample size. REM sleep Atonia Index was significantly higher in controls than in patients while the total number of chin movements/h during REM sleep was significantly increased in PD patients.

The top panel of Fig. 1 depicts the distribution histograms of the REM sleep mini-epoch amplitudes obtained in the two groups of

Table 1
Sleep architecture in control subjects and PD patients.

	Controls (n = 19)		PD patients (n = 27)		Mann–Whitney	Effect size
	Mean	S.D.	Mean	S.D.	p<	Cohen d
TIB (min)	502.5	55.92	454.8	62.68	0.05	0.803
SPT (min)	484.2	63.96	413.5	74.97	0.015	1.015
TST (min)	389.8	38.80	286.1	76.71	0.00028	1.706
Sleep latency (min)	14.5	11.99	28.1	36.87	NS	−0.496
REM latency (min)	109.3	107.81	153.3	89.16	NS	−0.445
SS/h	13.3	3.78	14.2	6.75	NS	−0.165
AWN/h	7.3	2.96	8.3	5.26	NS	−0.234
SE (%)	77.9	6.03	63.2	16.41	0.007	1.189
WASO (%)	18.8	8.04	30.6	15.66	0.036	−0.948
S1 (%)	6.8	3.65	9.5	6.77	NS	−0.496
S2 (%)	47.0	10.76	37.4	12.89	0.045	0.809
SWS (%)	14.1	7.46	12.1	8.25	NS	0.254
REM (%)	13.3	2.22	10.3	5.47	NS	0.719
REM sleep Atonia Index	0.933	0.037	0.751	0.240	0.0165	1.059
REM chin movements, number/h	59.7	38.23	177.5	156.80	0.0041	−1.032
PLMS index, number/h	9.2	8.92	13.5	29.69	NS	−0.200

TIB = time in bed; SPT = sleep period time; TST = total sleep time; SS/h = stage shifts per hour; AWN/h = awakenings per hour; SE = sleep efficiency; WASO = wakefulness after sleep onset; S1, S2 = sleep stages 1 and 2; SWS = slow wave sleep; REM = rapid eye movement sleep.

subjects. It is clear that controls tended to show a higher number of mini-epochs with amplitudes contributing to REM atonia (left, first column) while mini-epochs contributing to RSWA (columns on the right side of the second column) tended to be higher in PD patients.

3.2. Comparison between PD patients without or with RBD

We found that 16 (10 males and six females) out of our 27 PD patients were affected by RBD (PD + RBD) and 11 (seven males and four females) were not (PD − RBD). The upper part of Table 2 reports the comparison of some clinical features between these two PD subgroups. PD + RBD showed a significantly higher H&Y disease stage and took a significantly higher LED; also, disease duration tended to be longer in these patients and their MMSE tended to be lower, although these latter differences failed to reach statistical significance. None of the classical sleep structure parameters, as well as the PLMS index, showed statistically significant differences between the two PD subgroups and effect sizes were small in all instances but one (number of stage shifts), where it was only moderate. REM sleep Atonia Index was significantly lower in PD + RBD patients and the total number of REM sleep chin movements/h was significantly increased in PD + RBD patients. These differences were also evident when the comparison was made between the patients recruited from each centre (Troina: PD − RBD n = 5, mean Atonia Index 0.952 ± 0.036 S.D., PD + RBD n = 12, mean Atonia Index 0.612 ± 0.215 S.D.; Bologna PD − RBD n = 6, mean Atonia Index 0.942 ± 0.0656 S.D., PD + RBD n = 4, mean Atonia Index 0.631 ± 0.287 S.D.) and there was no significant difference between the centers.

The bottom panel of Fig. 1 shows the distribution histograms of the REM sleep mini-epoch amplitudes obtained in the two subgroups of patients. PD − RBD patient clearly showed a higher percentage of atonia mini-epochs (first column) while the proportion of non-atonia mini-epochs (columns 3–20) was clearly higher in PD + RBD patients.

Fig. 2 shows the REM sleep Atonia Index for each subject belonging to the control group or to the PD patient subgroups together with group average values and corresponding 95% confidence intervals. Following our recently published cut-off values for the REM sleep Atonia Index [5], the background of the figure is subdivided into three areas (Atonia Index < 0.8, 0.8 < Atonia Index < 0.9, and Atonia Index > 0.9) corresponding to clearly abnormal, moderately abnormal, and clearly normal values, respectively. It is possible to note that three out of the 19 controls presented moderately abnormal values while the remaining 16 showed REM sleep Atonia Index above 0.9; only one PD − RBD patient had a moderately abnormal REM sleep Atonia Index and all others showed values above 0.9; finally, only one PD + RBD patient had REM sleep Atonia Index above 0.9, four showed moderately abnormal values, and the remaining 11 showed clearly abnormal REM sleep Atonia Indices below 0.8. These differences were highly significant (chi-square test). Finally, considering as normal Atonia Index values >0.9, we calculated its sensitivity (0.938), specificity (0.909), positive predictive value (0.937), and negative predictive value (0.909) for the diagnosis of RBD in our PD patients (i.e., PD + RBD vs. PD − RBD).

4. Discussion

Under the hypothesis that RBD is characterized by specific chin EMG patterns in different clinical conditions, such as idiopathic RBD, multiple system atrophy (MSA), and narcolepsy [4,5,8], we have now expanded this range of conditions and have carried out the first study that analyzed in a strictly quantitative way the amplitude of the chin EMG during REM sleep in a well defined group of patients with PD. Although previous studies did include PD + RBD patients in their quantitative analysis of REM EMG, they either did not report separate results for this group, pooling them together with those of patients affected by other conditions associated to RBD, or compared PD + RBD to healthy controls subjects only [11,21]. The most important result of our study is that the REM sleep Atonia Index can detect the presence of clinically diagnosed RBD in PD patients with high sensitivity and specificity; this suggests that the REM sleep Atonia Index can be proposed as a valid support for the diagnosis of RBD in PD patients, particularly in dubious cases; this is bearing in mind, however, that we computed the Atonia Index in the submental muscle only. Occasionally, RBD patients may show normal REM atonia in the cranially innervated muscles, while loss of atonia is observed for other muscle groups.

The REM Atonia Index has been validated in the first paper published on this quantitative method [4] where we also reported the correlation coefficient between REM Atonia Index and the proportion of the total REM epochs scored as atonic according to the visual analysis by Lapierre and Montplaisir [6]. In four groups of subjects (young and old controls, idiopathic RBD and MSA patients), correlations ranged between 0.745 and 0.963 and were always highly significant. With the present study we add to that the evaluation of sensitivity and specificity of the Atonia Index in identifying the presence of RBD in patients with PD.

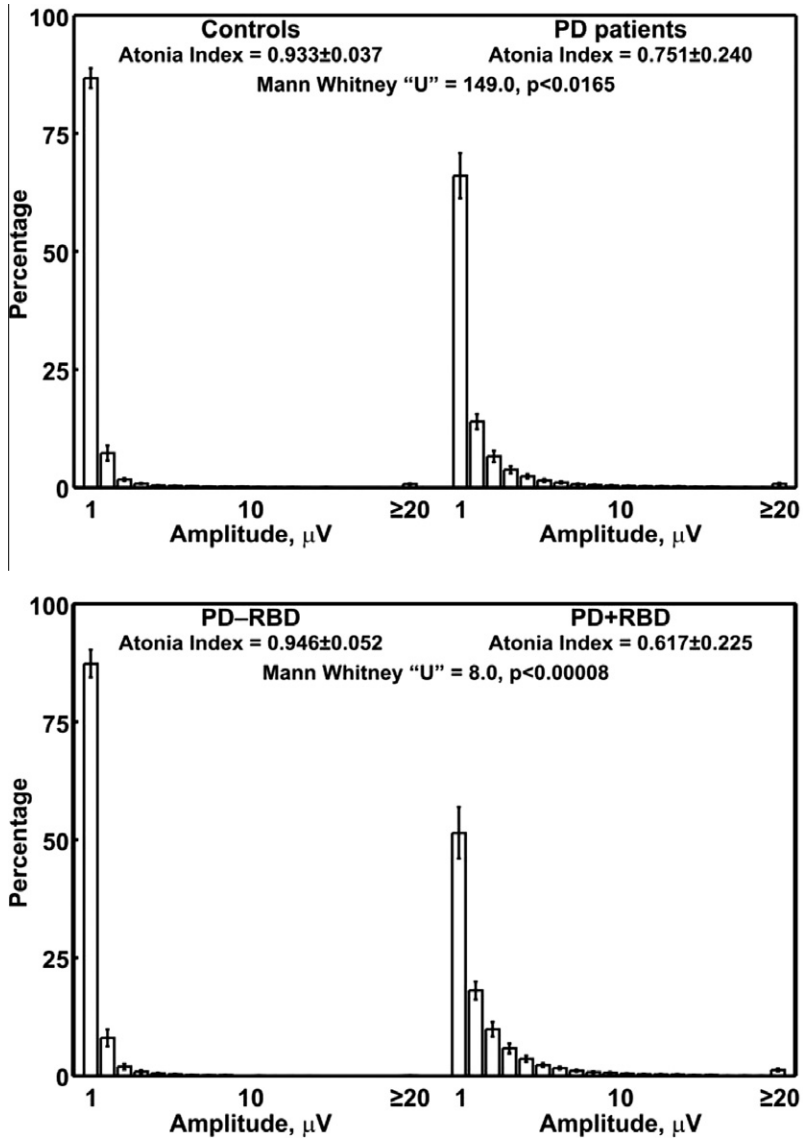

Fig. 1. Top panel: distribution histograms of mini-epochs amplitude for controls and PD patients. Bottom panel: distribution histograms of mini-epochs amplitude for PD patients without (PD − RBD) or with RBD (PD + RBD). The percentage of values in 20 amplitude (amp) classes is shown (1 = amp ⩽ 1, 2 = 1 < amp ⩽ 2, . . . , 19 = 18 < amp ⩽ 19, 20 = amp ⩾ 20 μV). All values are shown as means and standard errors (whiskers).

We would like to point out that the original description of the method by Lapierre and Montplaisir [6] was based only on five patients and five controls and only recently have Montplaisir et al. [22] reported results from 80 RBD patients compared to 80 controls. In our first paper [4], we reported 21 RBD patients, 10 MSA patients, and 34 controls; in our second paper employing this method [8] we reported 34 patients with narcolepsy (17 with and 17 without RBD); in the third article [5] we described a total of 89 subjects (patients and controls); and in our most recent publication [7] we have recruited 88 normal controls. If we add the 27 PD patients included in the study under evaluation (plus 19 controls) we reach a significant total number of recordings already analyzed with clear-cut results, normative data, and pathological values that are certainly not less validated than the method by Lapierre and Montplaisir [6,22].

In this study, PD − RBD patients presented with an REM EMG amplitude pattern that was consistently normal; this is clearly different from the results obtained in narcoleptic individuals, most of whom showed variable involvement of the chin EMG parameters, more evident in those with clinically evident RBD but present also in those without, even if to a smaller extent [8]. On the other hand, compared to MSA patients [4], the degree of loss of atonia was less pronounced in PD + RBD patients. This can be considered as indirect evidence supporting our original hypothesis that RBD is characterized by specific chin EMG patterns in different clinical conditions.

It is important to note that almost all patients included in this study were under dopaminergic therapy. Most probably, this is why we found PLMS levels similar to those of controls and in line with the amounts expected for their age [23]. We also included in this study three patients who had symptoms of RLS – not a rare event in PD [24,25] – but which had been controlled by the dopaminergic therapy. Nevertheless, our data suggest that dopaminergic therapy has virtually no effect on RBD. During REM sleep several mechanisms seem to determine and modulate the signal outflow to skeletal muscles, including increased glycinergic and GABAA-mediated inhibition [26,27], decreased noradrenergic and serotonergic excitation [28,29], and hypocretinergic state-dependent modulation [30]. Thus, REM atonia is most probably not mediated by a single or leading pathway, but rather by a complex network including many different neurotransmitter mechanisms [1,31]. In agreement with this view, our data seem to support the

Table 2
Clinical features and sleep architecture parameters in PD patients without (PD − RBD) or with RBD (PD + RBD).

	PD − RBD (n = 11)		PD + RBD (n = 16)		Mann–Whitney	Effect size
	Mean	S.D.	Mean	S.D.	p<	Cohen d
Males/females ratio	7/4		10/6		NS*	
Age, years	66.5	6.38	68.9	8.16	NS	−0.328
LED	232.7	197.03	565.9	427.39	0.038	−1.001
Hoehn & Yahr stage	1.5	0.52	2.0	0.74	0.048	−0.782
PD duration (years)	3.0	2.98	5.3	4.22	NS	−0.630
MMSE	28.6	1.35	26.9	3.84	NS	0.591
TIB (min)	440.1	53.64	464.9	68.00	NS	−0.405
SPT (min)	394.7	78.82	426.4	71.87	NS	−0.420
TST (min)	282.4	86.00	288.6	72.47	NS	−0.078
Sleep latency (min)	35.7	56.48	22.8	12.51	NS	0.315
REM latency (min)	163.8	100.24	146.0	83.33	NS	0.193
SS/h	16.2	4.54	12.9	7.78	NS	0.518
AWN/h	9.0	4.19	7.8	5.97	NS	0.233
SE (%)	64.2	17.51	62.6	16.16	NS	0.095
WASO (%)	29.4	12.61	31.5	17.82	NS	−0.136
S1 (%)	10.9	6.25	8.5	7.13	NS	0.358
S2 (%)	36.1	9.32	38.2	15.11	NS	−0.167
SWS (%)	12.9	7.08	11.6	9.15	NS	0.159
REM (%)	10.6	6.31	10.1	5.02	NS	0.088
REM sleep Atonia Index	0.946	0.052	0.617	0.225	0.00008	2.022
REM chin movements, number/h	52.9	39.16	263.1	149.63	0.00004	−1.922
PLMS index, number/h	12.7	19.78	14.1	35.58	NS	−0.051

LED = *L*-dopa equivalent dose; MMSE = mini-mental state evaluation; TIB = time in bed; SPT = sleep period time; TST = total sleep time; SS/h = stage shifts per hour; AWN/h = awakenings per hour; SE = sleep efficiency; WASO = wakefulness after sleep onset; S1, S2 = sleep stages 1 and 2; SWS = slow wave sleep; REM = rapid eye movement sleep.
* Chi-square test.

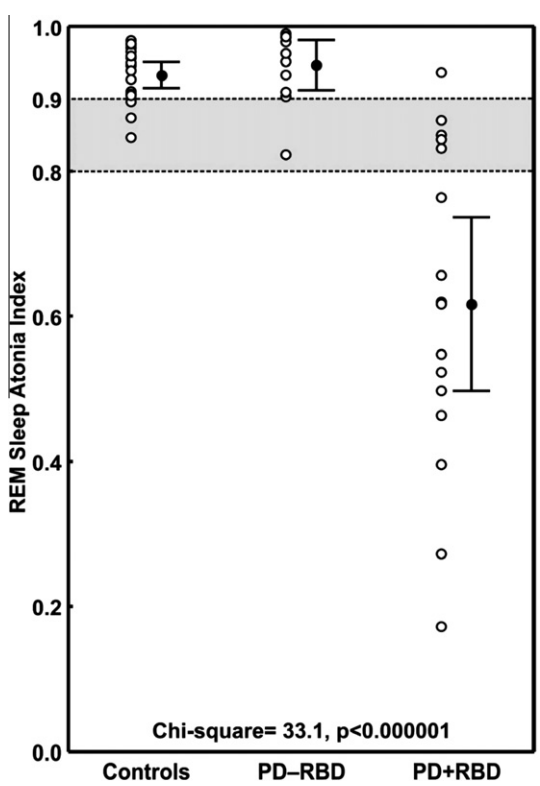

Fig. 2. Distribution of the REM sleep Atonia Index in each subject (empty circles) belonging to the control group or to PD patients without (PD − RBD) or with RBD (PD + RBD). Group average values are also shown with 95% confidence intervals (black-filled circles and whiskers).

idea that dopamine has little or no role in RBD [32,33] and do not support earlier studies indicating such a possibility [34,35].

Previous studies have reported that PD + RBD as compared to PD − RBD have longer disease duration [36–39], more severe motor impairment [40], and higher doses of dopaminergic agents [37,38,41]. Importantly, a recent large-scaled study of RBD in PD with video-supported polysomnography in all patients confirmed that RBD in PD is associated with longer disease duration, higher H&Y stages, and higher doses of levodopa [10]. In our sample we also found more severe motor impairment and higher doses of dopaminergic agents in PD + RBD.

It is also important to note that none of the controls nor the PD − RBD patient who had REM sleep Atonia Index between 0.8 and 0.9 presented dream enacting behaviors; in one of our previous studies [5] we reported a small group of RBD patients under therapy with clonazepam that still presented low Atonia Index values, although the clinical effect was reported to be good. This could indicate that this treatment may exert a clear effect on dream enacting behaviors, but may not be equally effective in reducing the Atonia Index. This needs to be confirmed in larger controlled studies but seems to be in agreement with the proposition of a different mechanism for dream enactment and RSWA [42].

Some potential limitations of the present study must be acknowledged. As already mentioned, the sample size was rather small, precluding more detailed analyses within subgroups and possibly resulting in low statistical power. To partly overcome this limitation we reported effect sizes in addition to the traditional test statistics. Overall, these indicated that we have not missed any large difference between the groups. In addition, RBD was carefully diagnosed according to standard criteria [3], including documentation of RSWA and complex behaviors, but we did not determine RBD severity. Although several measures to characterize severity of RBD have been published [43–47], there is currently no internationally established severity measure. In addition, only patients with mild and moderate PD were recruited; thus, our results might not apply to the entire range of PD severity. Finally, we only characterized REM chin EMG patterns and did not include tibialis anterior or other muscles. In our previous studies [4,5,8] we found that chin EMG patterns clearly distinguished between healthy controls and idiopathic RBD, MSA, or narcolepsy. In these studies and our aim was to analyze PD and PD + RBD with the same set

of parameters, and the parameters were also in agreement with a visual quantification of phasic EMG activity in PD and RBD patients, showing similar levels of abnormality in the two conditions [48]. Whether the additional consideration of other muscle groups during REM sleep will add to this distinction will have to be explored in future studies.

In summary, we analyzed REM chin EMG patterns in healthy controls and patients with PD with and without RBD. We found that the REM sleep Atonia Index clearly differentiated between PD with RBD and PD without RBD with high sensitivity and specificity, recommending this index as an objective measure to support and aid the diagnosis of RBD in PD.

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Conflict of Interest

The ICMJE Uniform Disclosure Form for Potential Conflicts of Interest associated with this article can be viewed by clicking on the following link: doi:10.1016/j.sleep.2012.01.003.

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